Assets, Rents, and the Socialized Buildout

An accounting of the AI capital cycle: market values, capital stocks, revenue requirements, and the telecom precedent

Prepared by Claude (Anthropic) from a dialogue with Matthew Tomei · July 27, 2026

Abstract. Roughly $27 trillion of equity value has attached to AI-related companies since November 2022, against which the physically deployed, depreciation-adjusted AI capital stock is on the order of $1.2 trillion. This report separates the two quantities and sizes what must fill the gap. It builds a vintage model of the AI capital stock, reprices it at competitive supply-chain margins, backs out the economic-rent stream the market has capitalized, and inverts that requirement into the share of world GDP that must flow to AI at plausible margins. It then calibrates the requirement against forty years of IT-spending history, develops the telecom buildout of 1996-2002 as the structural precedent - a covertly socialized infrastructure program whose losses funded a commons - and maps the vendor-financing pathologies of that cycle (Lucent, Nortel, Winstar) onto today's circular deal graph. Reported figures and estimates are distinguished throughout; sources and model assumptions appear at the end.

1. The Question and a Correction

The prompt for this analysis was a Goldman Sachs Insights piece (July 2026) asking whether US equity valuations have outrun fundamentals [1]. Its load-bearing figures: AI-related companies have added roughly $27 trillion of market value since November 2022 (up from about $19 trillion seven months earlier), while the bank's baseline estimate of the present discounted value of AI-related capital revenues to US companies is roughly $9 trillion. US technology investment as a share of GDP has surpassed its late-1990s peak, and the 2026 spending plans of the largest cloud and computing companies came in about fifty percent higher than estimates made only six months prior. Goldman's stated risk is that the market overestimates the persistence of the earnings streams - particularly for the companies supplying the capex boom - beyond a two-to-three-year horizon, and it concedes the reflexive point that the investment boom itself currently generates a substantial portion of the profits justifying the prices.

One correction frames everything downstream: the $27 trillion is not investment. It is market-capitalization appreciation - capitalized expectation. Actual capital expenditure is an order of magnitude smaller. The four largest hyperscalers spent about $226 billion in 2024 and about $410 billion in 2025, and after Q1 2026 earnings guided to roughly $725 billion combined for 2026, with Microsoft alone at $190 billion [13, 15]. Goldman projects $5.3 trillion of big-four capex over 2025-2030 [1]. Cumulative AI-attributable investment since ChatGPT, built up carefully in Section 3, is roughly $1.0-1.2 trillion gross. Realized investment is therefore about six percent of the market-value move; the other ninety-four percent is expectation. That ratio is the article's tension stated as arithmetic, and the remainder of this report is an attempt to give each side of the ratio its correct accounting.

2. The Capex Boom and Its Persistence

The mechanical core of the persistence concern is that supplier revenue is a derivative of the capex flow, not of the installed stock. If spending merely plateaus, supplier growth goes to zero; if it reverts toward replacement level, supplier revenue contracts outright. The historical base rate is unkind - US telecom capex roughly halved between 2000 and 2003 - but the depreciation structure here genuinely differs from fiber. Dark fiber carried near-zero replacement demand; accelerators on four-to-six-year lives at a ~$500 billion annual run rate imply a replacement floor above $100 billion per year and rising with the installed base. Short asset lives are bearish for the buyers' balance sheets and bullish for recurring supplier revenue: the same fact, booked on opposite sides.

The stronger evidence for the limited-time view is the financing signature. Capital intensity now runs 45-57 percent of revenue for the big four and 86 percent for Oracle; Amazon's free cash flow is projected to turn negative; roughly $108 billion of debt was raised by hyperscalers in 2025 with on the order of $1.5 trillion projected over the coming years [15]. Migration from cash-funded to debt-and-lease-funded spending is the canonical late-phase marker: it extends the boom past internal cash generation and makes the flow fragile to sentiment.

2.1 Production capacity: reallocation, not new build

A second-order question with first-order consequences: how much of the apparent explosion in compute production is new physical capacity versus existing capacity redirected and repriced? So far, mostly the latter. Leading-edge logic barely grew in aggregate: high-performance computing reached 61 percent of TSMC revenue by Q1 2026, up from roughly a third before the boom - the same fabs, shifted from phones to accelerators as mobile demand softened [14]. Memory is the purest relabeling: HBM consumed 23 percent of global DRAM wafer starts in 2026 versus 8 percent in 2024, at roughly a three-to-one wafer-intensity penalty against DDR5; Micron exited consumer memory entirely, and DDR5 contract pricing more than doubled [16]. The one place physical capacity genuinely multiplied is advanced packaging, because it was the binding constraint and is cheap relative to fabs: TSMC CoWoS went from about 35,000 wafers per month at end-2024 to about 75,000 at end-2025, targeting 125,000-130,000 by end-2026 - a near-4x expansion that still trails demand by an estimated 10-20 percent [16]. Meanwhile TSMC's entire 2026 capital budget is $52-56 billion and total semiconductor-industry capex runs about $200 billion per year - against more than $725 billion of downstream datacenter spending. The production base for compute is expanding far more slowly than spending on its output.

That asymmetry concentrates pricing power upstream - TSMC at a 66.2 percent gross margin, memory prices doubling - and Microsoft attributed $25 billion of its 2026 capex increase purely to rising memory and component costs [13, 14]. A nontrivial slice of measured capex growth is therefore intra-supply-chain price inflation: scarcity rents, which are definitionally transient because they persist only while supply lags. The market capitalizing scarcity rents as durable franchise earnings is the mechanically precise form of Goldman's earnings-bubble worry. The watch variables are the CoWoS supply gap narrowing toward ~10 percent through 2026 and the new memory fabs landing in 2027-28 - fresh supply arriving just as capex growth must decelerate is the classic semiconductor bullwhip setup. The falsifier runs through end demand: cloud AI revenue compounding at rates like Google Cloud's reported 63 percent year-over-year is the kind of number that, sustained, absorbs the supply as it lands [13].

3. An Asset-Side Estimate: The AI Capital Stock

The direct route around speculative pricing is to build the asset side: a replacement-cost capital stock, compared against market value - Tobin's Q at sector level. Q above one is not per se irrational, since market value legitimately capitalizes rents on future investment as well as installed assets; the exercise does not eliminate the speculative component but quarantines it into a single residual that can be sized and argued about. Aggregate versions exist - Econbrowser ran whole-market q against the intellectual-property capital stock in December 2025 [17] - and Goldman's $9 trillion is the profit-side dual, but an AI-sector decomposition does not appear to have been published. We construct it here.

3.1 Vintage build-up

Global AI-attributable gross capex by vintage, estimated from hyperscaler disclosures (applying an AI share rising from roughly 30 percent in 2023 to 75 percent in 2026, per CreditSights [15]) plus Oracle, the neoclouds, xAI and OpenAI's direct spending, Chinese platforms, and sovereign programs:

Vintage	Low ($B)	Central ($B)	High ($B)	Age at mid-2026 (yr)
2023	45	65	85	3.0
2024	150	185	215	2.0
2025	400	460	500	1.0
H1 2026	310	370	420	0.25
Cumulative gross	905	1,080	1,220	-

Table 1. Vintage gross AI-attributable capex (author's estimates from reported figures).

A supply-side cross-check anchors the range. Nvidia's datacenter revenue was $115.2 billion in FY2025, $193.7 billion in FY2026, and $75.2 billion in Q1 FY2027 alone - cumulative roughly $430 billion from early 2023 through April 2026, approaching $500 billion through July [12]. At Nvidia around 60 percent of AI IT-equipment value (custom accelerators such as TPU and Trainium at cost, AMD, non-Nvidia networking making up the rest), implied IT spending of $750-830 billion pushes the true gross figure toward the upper half of the modeled range.

3.2 Depreciation, add-ons, and the competitive-cost adjustment

Depreciating by asset class - 62 percent IT equipment on five-year straight line, 23 percent power and cooling infrastructure on fifteen, 15 percent shell and land on thirty, mid-year convention - removes roughly $160 billion. To the net datacenter stock we add the AI-allocated share of upstream fab plant (TSMC and the memory makers, ~$125B), AI-attributable grid and power investment (~$65B), and model intellectual property at cost, excluding compute already counted (~$55B). One further adjustment matters: book value embeds the scarcity rents, because the accelerator stock was purchased at ~75 percent gross margins and memory at doubled prices. Repricing the IT stock at competitive supply-chain margins cuts it roughly in half, giving the long-run equilibrium anchor that entry drives toward - and implying the asset base itself deflates by roughly $250 billion if supplier pricing normalizes.

Case	Gross capex	Net DC stock	Total net stock*	Competitive-cost stock
Low	905	773	958	746
Central	1,080	919	1,164	911
High	1,220	1,034	1,339	1,056

*Table 2. AI capital stock at mid-2026, $B. *Includes upstream fabs, power/grid, and model IP at cost. Author's model; assumptions in Appendix A.*

3.3 The rent decomposition

Against the $27 trillion market-value gain - or roughly $18 trillion taking Goldman's own caveat that perhaps a third reflects non-AI businesses - sector Q runs 15-23x. The rent capitalization (market value minus assets) is about $17 trillion on the conservative attribution. Backed out as a required flow at an 8 percent discount rate:

AI-attributable MV	Rent capitalization	Growing 3%/yr	Flat perpetuity	Decaying 10%/yr
$14T	$12.8T	$0.64T/yr	$1.03T/yr	$2.31T/yr
$18T	$16.8T	$0.84T/yr	$1.35T/yr	$3.03T/yr
$27T	$25.8T	$1.29T/yr	$2.07T/yr	$4.65T/yr

Table 3. Required economic-rent flow implied by rent capitalization at r = 8%, central asset stock of $1.18T. Author's model.

Current realized AI economic rents run roughly $330-350 billion per year: Nvidia annualizing about $200 billion of operating income, TSMC's AI-attributable share around $40 billion, memory scarcity profits around $60 billion, thin hyperscaler direct-AI margin, and an application layer near zero [12, 14]. The market therefore requires the rent stream to grow roughly 2.5-4x and persist indefinitely - while 85-90 percent of today's rents sit in the supply chain, in exactly the scarcity-rent form that mean-reverts. The reconciliation with Goldman is clean: $1.2 trillion of assets plus their $9 trillion rent PDV accounts for about $10 trillion against $18 trillion of AI-attributable market gain, leaving roughly $8 trillion as the speculative residual under their own baseline. A fully auditable variant exists for anyone who wants it: summing net property, plant and equipment plus finance-lease assets from the 10-Ks of the AI complex and taking the delta since Q4 2022 yields roughly $450-500 billion for the big four through 2025, corroborating the vintage build once non-AI growth is stripped and the rest of the ecosystem added.

4. The Revenue Requirement: What Share of GDP Must Flow to AI

Inverting the rent requirement into a revenue requirement demands one construction choice: AI revenue must be measured as external revenue - sales from the consolidated AI stack to the rest of the economy. Summing layer revenues double-counts (Nvidia's revenue is Microsoft's cost is OpenAI's cost), and today most of the stack's profit is funded by intra-stack capex - by capital markets rather than final demand. The transition being priced is precisely that external revenue replaces the capex flow as the funding source of the rents.

Required after-tax profit by roughly 2030 is the rent stream plus normal returns on what will by then be a ~$5 trillion capital stock: about $1.25 trillion per year in the growing-rents case, $1.75 trillion flat, $3.4 trillion if rents decay. Dividing by consolidated net margin on external revenue:

Case (required profit)	30% margin	20% margin	12% margin
Growing rents ($1.25T/yr)	$4.1T = 3.1% W / 12% US	$6.2T = 4.6% W / 18% US	$10.4T = 7.7% W / 30% US
Flat perpetuity ($1.75T/yr)	$5.8T = 4.3% W / 17% US	$8.8T = 6.5% W / 25% US	$14.6T = 10.8% W / 42% US
Decaying rents ($3.4T/yr)	$11.4T = 8.5% W	$17.2T = 12.7% W	$28.6T = 21.2% W

Table 4. Required external AI revenue and share of 2030 world (~$135T) and US (~$35T) GDP. Author's model; Appendix B.

Three readings. First, the decay row is the valuation restated as a theorem: the pricing requires durable moats, because without them no plausible GDP share closes the gap. Second, the US-only column is arithmetic nonsense at 12-42 percent of GDP, so the valuations mathematically presuppose global revenue capture - which sits awkwardly against export controls and Chinese self-supply. Third, the bracket: the floor - merely making the buildout NPV-neutral, covering depreciation (~$940B/yr on the 2030 stock, dominated by five-year IT lives), operations, and cost of capital with zero rents - is about $1.7 trillion per year, 1.3 percent of world GDP. Current external AI revenue (cloud AI services, the application layer, enterprise on-premise; author's estimate) runs perhaps $300-400 billion, about 0.3 percent of world GDP. The sector needs roughly 4-5x revenue growth to stop destroying value and 12-18x to justify the market pricing.

For scale: 3-6.5 percent of world GDP means external AI spending matching or exceeding all current global IT spending and two to three times all software plus cloud combined, and $1.25-1.75 trillion of after-tax profit stands against a total US corporate-profit pool of roughly $3.9 trillion. Which raises the question the GDP framing usually obscures: which spending pool funds this? IT budgets cannot - they total about five percent of world GDP and AI will not take all of them. Consumer discretionary does not produce trillions at software margins. The only pool of sufficient size is labor compensation, roughly 53 percent of GDP, about $70 trillion by 2030. If AI vendors capture around 30 percent of the labor-cost savings they create, then $4-7 trillion of revenue implies AI substantively addressing 19-33 percent of global labor income. The percentage-of-GDP assumption thus decomposes into four separately contestable conjuncts: partial automation of a fifth to a third of world labor; the AI complex capturing about 30 percent of that surplus rather than competing it away to users; doing so at 25-30 percent consolidated margins; and moats durable enough that the rents do not decay. The margin assumption and the moat assumption are the same assumption in different clothes. Telecom built the internet at ten percent margins and never earned its cost of capital; software is the other precedent. The market has priced the software outcome.

5. Historical Calibration: IT Spending and GDP

Gartner's April 2026 revision puts worldwide IT spending at $6.31 trillion for 2026, up 13.5 percent - about 5.3 percent of world GDP (~$120T). Hardware is $788 billion of datacenter systems plus $856 billion of devices, about $1.64 trillion or 1.4 percent of world GDP; software runs $1.44 trillion, IT services $1.87 trillion, communications services $1.36 trillion [2].

Segment	2025 ($B)	2025 growth	2026 ($B)	2026 growth
Data center systems	489	+46.8%	788	+55.8%
Devices	783	+8.4%	856	+8.2%
Software	1,244	+11.9%	1,440	+15.1%
IT services	1,719	+6.5%	1,870	+8.7%
Communications services	1,304	+3.8%	1,360	+4.8%
Total	5,540	+10.0%	6,310	+13.5%

Table 5. Worldwide IT spending by segment. 2025 figures and 2025 growth from Gartner's October 2025 release [3]; 2026 from the April 2026 revision [2] (2026 devices growth as reported against revised 2025 base).

The historical shape is more interesting than a simple rise, and it disciplines the Section 4 requirement. The global ratio has essentially round-tripped: roughly 5.3 percent of world GDP around 2008, drifting down to about 4.3 percent by 2019-20 as communications stagnated in absolute terms while GDP grew, then inflecting back - 4.8 percent in 2025, 5.3 percent in 2026 (historical shares are author's estimates from Gartner series and IMF GDP data). The headline share sat in a 4.3-5.3 percent band for two decades, and 2026 is the first decisive break above the band's ceiling. Composition rotated completely underneath: telecom fell from nearly half the total to about a fifth; software went from roughly $300 billion in 2010 to $1.44 trillion; and datacenter systems - the line that matters for this report - sat at $150-200 billion for essentially the entire 2010s, four to five percent of IT spending, before going roughly $243B (2023) to $333B (2024) to $489B (2025) to $788B (2026): a tripling in three years, now 12.5 percent of all IT.

The mechanism behind two decades of flatness is worth naming: compute consumption grew exponentially throughout, but hardware price deflation - Moore's law plus competitive supply - absorbed it, holding the nominal line flat. The current breakout is partly the suspension of that deflation: accelerators priced at 75 percent gross margins and HBM at doubled prices mean the datacenter line now grows with volume and price together, and Gartner's own commentary attributes its upward revisions substantially to memory pricing [2]. Some fraction of the IT share of GDP finally rising is the scarcity-rent regime showing up in category data. As a base rate for Section 4, this is sobering: the share needed forty years to reach five percent and then did not grow for twenty, and the valuation math needs a second IT-sector-sized wedge - 3-6.5 percent of world GDP - inside a decade. The one measurement caveat cutting the other way: Gartner's perimeter excludes the ad-funded consumer internet (Google's and Meta's roughly $700 billion of revenue is not counted as IT spending), and the BEA's broader digital-economy account puts the fuller US tech share near ten percent of GDP, up a couple of points over two decades. The economy can grow a multi-point tech wedge; it did so once, over about twenty-five years, funded largely by advertising's fixed share of GDP. The open question is whether labor substitution funds the second one faster.

6. The Telecom Precedent: Capital That Never Returned

The claim that the sector which built the internet never earned its cost of capital has a real empirical record. Between the 1996 Telecommunications Act and 2002 the sector raised on the order of $1.5-2 trillion in equity and debt and roughly tripled annual capex, peaking around $110-120 billion per year in the US in 2000 before collapsing by half within three years. The demand forecast underneath - internet traffic doubling every hundred days, the WorldCom/UUNET claim - was off by roughly an order of magnitude (actual doubling was close to annual; Odlyzko spent years dismantling the meme [18]), while DWDM multiplied capacity per fiber strand roughly a hundredfold mid-buildout. By 2002 an estimated 3-5 percent of installed long-haul fiber was lit and bandwidth prices on major routes had fallen more than ninety percent. WorldCom - $107 billion of stated assets, $11 billion of fraudulent accounting - became the largest bankruptcy in US history to that date; Global Crossing spent roughly $12-15 billion building its network and sold control out of bankruptcy for $250 million, about two cents on the dollar; Level 3 survived a 99 percent drawdown. European carriers paid about EUR 110 billion for 3G spectrum in 2000, mostly written off - a direct shareholder-to-treasury transfer. The Economist's 2002 tally ran to roughly $2 trillion of erased market value and $1 trillion of debt [21]. Nor was it only the bubble cohort: sector-level studies through the 2000s and 2010s consistently put telecom returns on invested capital at 5-8 percent against 7-9 percent costs of capital. Persistently, on both sides of the Atlantic, the industry operating the internet's transport layer earned less than its capital cost.

6.1 The productive-bubble thesis

From a global rather than shareholder perspective, this admits a stronger reading: the internet's physical layer was built with investor money that never saw a return - a covertly socialized infrastructure program, neither consensual nor centrally planned, but effective. Nordhaus's estimate is the clean theorem behind it: even in successful innovation, producers capture on the order of 2.2 percent of the social surplus they create [19]. A productive bubble is the degenerate case where private capture goes negative - investors pre-fund infrastructure whose surplus they cannot appropriate, and the crash is the moment that fact is marked to market. This is Perez's installation/deployment structure and Janeway's productive bubble [20]: the crash transfers assets to new owners at written-down cost basis, and that basis is what makes deployment-era services cheap. Google bought dark fiber from bankrupt carriers for pennies mid-decade; YouTube launched in 2005 into a bandwidth regime ninety percent cheaper than any rationally priced buildout would have produced. The railway mania is the same play run earlier: British railway investment reached roughly 5-7 percent of GDP in 1847, shareholders earned below consol yields on risky capital, and Britain received a national network it would never have voted to fund through taxation. Because the losses were equity-financed and diffuse - pensions and retail absorbed them - the 2001 recession was mild; bank-funded buildout busts (1873, 2008) propagate instead of dissipating.

Two complications keep the frame honest. First, within the investor class it was a transfer, not a shared sacrifice: insiders at Qwest and Global Crossing sold billions near the top, conflicted research and the IPO machinery pulled late retail money in, and $11 billion of WorldCom's numbers were fabricated. Consensual socialization it was not. Second, Odlyzko's sharper point: the bubble funded the fundable layer, not the binding one. Long-haul transport was a small fraction of total network cost with easy entry, so speculative capital pooled there; the actual bottleneck - the last mile - was built later, slowly, by incumbents on regulated economics. Bubbles allocate by legibility and venture-shape, not by bottleneck. The AI equivalent of the last mile is plausibly power interconnection and enterprise integration, which is roughly where capital moves slowest today.

One asymmetry deserves heavy weight in any mapping to AI: the gift mechanism requires the capital stock to outlive the crash. Fiber in conduit had twenty-five-year life and near-zero carrying cost - it could sit dark waiting for YouTube. The AI stock modeled in Section 3 is 62 percent IT equipment on roughly five-year lives: an AI overbuild does not bequeath, it evaporates. The durable residue would be the shells and power infrastructure (roughly 38 percent of the stock), the trained workforce, and - the genuinely interesting analogue - open-weight models. And the financing drift cuts the other way: dot-com losses were equity-diffused, while this cycle's debt migration, private credit, and special-purpose vehicles move it toward the banked-bubble category, where socialization stops being benign and becomes literal - bailouts and ratepayers, the latter channel already active in utility bills.

7. The Second Harvest: What the Losses Actually Bought

Take the socialized-buildout claim at full strength and the retrospective justification of the telecom writeoff was never merely cheap video. The overbuilt network did two things no planner ordered. It induced humanity to transcribe itself - the web, the forums, the code repositories, trillions of tokens produced as exhaust by people using bandwidth priced below its buildout cost - and it grew the ad-funded engine that later paid for the laboratories. The corpus and the cash flows were both second harvests of the glut. On the longest accounting, the 2000-02 losses were the capital contribution to machine intelligence: privately negative, socially incalculable, with the payoff arriving twenty years late in a form nobody was pricing. That is not a metaphor stretched over the facts; it is the causal chain.

Open weights are then the piece that makes this cycle's version of the bequest structurally stronger than fiber. Fiber persisted because glass in conduit does not rot. Weights persist because of irreversibility plus zero-cost reproduction - and they add a property no prior infrastructure had: the artifact teaches its own manufacture. Every served token is a lesson; distillation means the product can be interrogated into yielding its replacement. Nordhaus found producers capture about two percent of innovation surplus on average; for this good that starts to look like a ceiling, because appropriability leaks through the product's own function. The physical form is maximally unpossessable - the crown jewels of a multi-trillion-dollar complex are a few hundred gigabytes, which fit in a backpack and therefore in no vault. Export controls govern chips because nobody can durably govern files. Two refinements keep the analogy honest. The dark fiber of this cycle is not the weights already published - those are lit and carrying traffic - but the elicitation overhang: capability resident in released artifacts that current scaffolding has not yet extracted. And the crash scenario is synergistic for the commons rather than destructive: a bust means fire-sale accelerators the way 2002 meant two-cent bandwidth, and written-down compute plus free weights is exactly the deployment-era cheapness the next platform gets built on. The rent capitalization of Section 3 is crash-exposed; the endowment is crash-hedged.

The friction is real and should be stated against the grain of the optimism: the gift is parasitic on the race. Open releases are strategic byproducts - complement commoditization, positioning, recruiting - funded by the very rent expectations sized above. The floor rises only while the frontier is contested; the day the funding stops, the last release freezes as the permanent public endowment, trailing closed capability by an unknowable gap. And diffusion through this commons selects on capability per dollar, not on wisdom - a discrimination problem with no filter at all, because the unpossessable is also the unrecallable. There is an old resonance here that requires no forcing: the provision that spoiled when hoarded, save the portion set aside; closed frontier value decays on roughly that schedule - months to obsolescence - while the released portion keeps. Property was the wrong category from the start; the balance sheets are merely the last to find out.

8. The Circular Economy: The Ouroboros Formalized

A recent Atlantic essay describes the AI economy as a trillion-dollar ouroboros of buying and selling, investment and equity staking, largely among a handful of firms - Big Tech advancing money to AI startups to buy cloud services from Big Tech [10]. Two underlying sources supply the microstructure for precisely the aggregate claims built above: the funding gap and the margin-stacking constraint.

8.1 The firm-level instance: OpenAI

PitchBook's analysis (Harrison Rolfes) is the firm-level version of the Section 3 inversion, and it closes consistently: to justify an $852 billion valuation, OpenAI must generate $95-105 billion of free cash flow by 2030 - roughly the same 8-12x forward-rent multiple the aggregate implied - while on Q1 trajectory it loses $10-30 billion that year instead [5]. (Free cash flow: operating cash flow minus capital expenditure - the cash actually left after running and investing in the business; FCF margin is that as a share of revenue.) Q1 2026 revenue was $5.7 billion at a negative 122 percent adjusted operating margin: $2.22 spent per dollar earned. Even OpenAI's own plan - roughly $280 billion of 2030 revenue, up from $13.1 billion in 2025, with its infrastructure target cut from $1.4 trillion to about $600 billion [11] - requires a ~36 percent FCF margin at scale, better than Microsoft's or Apple's mature cash economics, achieved within four years while buying compute at scale against Google, Anthropic, and free open weights. The infrastructure target itself is mostly not owned assets but long-term purchase commitments: named pledges of roughly $250B to Azure, $300B to Oracle, $138B to AWS, $22.4B to CoreWeave, $10B to Broadcom, up to $100B from Nvidia (still a letter of intent per Nvidia's CFO as of December 2025), plus a 6GW AMD deal carrying warrants for up to ten percent of that supplier [7, 8, 9]. Roughly $820 billion of commitments stand against a $25-33 billion current revenue run rate - and this is the designated demand terminus of the whole loop, the node where external revenue is supposed to enter, with 85 percent of its 800-900 million weekly users on free tiers [5, 6]. The February 2026 markdown of the target from $1.4T to $600B was therefore not housekeeping: the terminus wrote down its own pledged draw by roughly $800 billion, and those pledges are exactly the flows other layers had capitalized.

8.2 Consolidation, double-capitalization, and margin stacking

Draw the accounting perimeter around the complex and eliminate intra-group transactions, as consolidation would for any conglomerate: every circular deal vanishes in elimination entries, and what remains is the external P&L - roughly $300-400 billion of outside revenue against $700 billion to $1 trillion of cash out. That gap is the socialization flow, running now, continuously - the structural difference from telecom, which socialized at the writeoff. The funding legs are index flows (the largest seven technology companies' ~35 percent S&P 500 weight makes every passive retirement dollar a mechanical contribution), the debt migration, and sovereign capital. The Atlantic's what-happens-to-one-happens-to-all is then simply the observation that intra-complex claims net to zero: the complex is economically one consolidated entity whose equity tranches happen to trade separately.

The sharpest structural feature is double-capitalization of a single uncertain flow. The same expected OpenAI spending appears as Oracle's remaining performance obligations - RPO, the accounting disclosure of contracted-but-undelivered revenue, which exploded to roughly half a trillion dollars largely on that one contract and repriced Oracle's equity enormously - as Nvidia's backlog, as CoreWeave's contracted revenue, and as hyperscaler committed-spend disclosures. Each layer capitalizes its margin slice of the flow into its own market capitalization, while OpenAI's $852 billion capitalizes the end revenue that must fund all of them. A trillion dollars of commitments is not a trillion of demand; it is one ~$280-billion-per-year aspiration pledged severally, which is why tail correlation inside the perimeter approaches one. And the margin-stacking constraint is already binding: the April renegotiation capping Microsoft's revenue share at $38 billion through 2030 - without which, per PitchBook, positive free cash flow is not possible - is rent reallocation inside the loop, by private negotiation, before external revenue has arrived [5]. The layers' summed margin assumptions exceeded what the flow could fund, and the reconciliation has quietly begun.

8.3 Equity-linked vendor financing and the corrupted signal

The instrument set rhymes with 1999 but differs in one load-bearing way. Lucent and Nortel did vendor financing through receivables, concentrating customer credit risk on their own working capital. This cycle's version is equity-linked: supplier investments in customers, warrants granted to a customer, utilization backstops (Nvidia's arrangement to absorb unused CoreWeave capacity), compute credits structured as investment [7, 8, 9]. Equity-linked financing is system-safer - losses diffuse to shareholders rather than triggering a receivables death spiral - but epistemically worse, because it contaminates the demand signal itself. When the supplier funds the purchase and books it at 75 percent gross margin, and when Google's and Amazon's reported earnings include paper gains on their laboratory equity stakes (a point Fortune quantified in April 2026), the market's instruments for measuring whether external demand exists are themselves inside the loop. Price discovery degrades because nearly every transaction carries a financing leg. Add OpenAI's DeployCo joint venture guaranteeing private-equity investors a 17.5 percent annual return [9] - equity risk repackaged as a debt-like claim - and the marginal capital is no longer risk-bearing in Perez's sense at all; it is rented, which is how installation phases end. The IPO wave - OpenAI and Anthropic listing alongside Databricks [5] - is the 1999 rhyme: exposure migrating from strategic balance sheets to the index-holding public precisely as the internal reconciliation begins.

9. Vendor Financing, 1996-2002: Lucent, Nortel, Winstar

The precedent deserves its detail, because both protagonists were blue-chip institutions. Lucent - Bell Labs plus Western Electric, spun out of AT&T in 1996 - was briefly the most widely held stock in America, peaking near $250 billion of market value on about $38 billion of revenue. Nortel traced to 1895 and at its July 2000 peak represented roughly a third of the entire Toronto index at about C$400 billion. Both lost ~99 percent of their value within two years; Lucent merged into Alcatel in 2006 for $13.4 billion - a nineteenfold markdown from peak - and Nortel ceased to exist.

The mechanism: the 1996 Act spawned hundreds of competitive local exchange carriers - thinly capitalized startups whose business plan was borrowing money to buy network equipment - and the vendors competed for those orders by supplying the purchase money themselves. Lucent's vendor-financing commitments reached roughly $8 billion, Nortel's several billion more, the industry perhaps beyond $20 billion. The accounting made it lethal: a vendor-financed sale books as revenue immediately, at full margin, with a receivable on the asset side - so the income statement showed booming demand while the balance sheet quietly accumulated credit exposure to customers solvent only while capital markets stayed open. Demand was partially synthetic - the vendors' own capital returning as orders - and the street extrapolated the revenue growth without consolidating the loop. When CLEC funding closed in 2000-01, customers died, receivables became writeoffs, and the extrapolated demand evaporated simultaneously. The canonical case is Winstar: Lucent committed $2 billion of financing to a fixed-wireless CLEC; Winstar went bankrupt in April 2001 and sued Lucent for $10 billion for cutting off the credit line - the customer suing the supplier for refusing to keep funding its own purchases, the loop stated as a legal claim.

Each company added its own pathology. Lucent pulled demand forward with quarter-end discounts and distributor stuffing, restated $679 million of fiscal-2000 revenue, and ultimately faced SEC charges over roughly $1.1 billion of improperly recognized revenue [22]; the stock went from $84 to under a dollar, survival required spinning off Avaya and Agere and cutting headcount from ~130,000 toward ~35,000. The signature is the triple hit - revenue restated, receivables written off, credibility gone - each amplifying the others. Nortel's variant was paying about $32 billion in inflated stock for acquisitions and writing essentially all of it off - a $19.4 billion single-quarter loss in mid-2001, roughly $27 billion for the year - followed by a second act in which post-crash management released accounting reserves to manufacture a 2003 return to profitability, triggering executive bonuses, terminations for cause, two multi-year restatements, and criminal trials ending in acquittal in 2012 but never in recovered credibility. It filed in January 2009 - Canada's largest bankruptcy [23]. The estate sale connects to the residue thread: the operating businesses fetched a few billion in pieces while the single most valuable asset was the patent portfolio, sold for $4.5 billion to an Apple-Microsoft-Ericsson consortium that outbid Google, whose opening bid was literally $3.14159 billion - pi. The durable product of a century-old institution was pure intellectual property, purchased for exclusion. Cross-border litigation over splitting the estate burned roughly $2 billion in professional fees over eight years, and some twenty thousand pensioners took devastating losses.

What the detail adds to the AI mapping. First, the vendors acted knowingly, inside a prisoner's dilemma: if Lucent would not finance Winstar, Nortel or Cisco would, and land-grab logic made unilateral prudence equivalent to ceding share. The same structure operates now - each equity-linked deal defensible as positioning, the ensemble a machine for manufacturing the demand signal. Second, the failure mode differs by instrument: receivables financing put customer credit risk on the vendor's own working capital, so death came fast and triple-barreled; equity-linked losses mark to a line investors already treat as speculative, so an unwind would be slower and less mechanically self-reinforcing - genuinely safer plumbing around the same corrupted signal. Third, collateral: repossessed CLEC networks were worth nearly nothing to anyone else, whereas repossessed GPUs are fungible - though fungible into a glut, which is how two-cent bandwidth happened. Fourth, any unwind's complexity: Nortel's estate had one jurisdictional web to untangle and consumed $2 billion in fees; the OpenAI-Microsoft-Oracle-Nvidia-SoftBank claim graph would make that fight look like small claims. The cheerful coda stands: Winstar's stranded assets and the CLECs' dark fiber became the cheap substrate of the deployment era. The vendors financed a commons and were repaid in bankruptcy claims.

10. Synthesis and Watch Variables

The through-line of this report is a single decomposition applied at every scale. Market value equals replacement-cost assets plus capitalized rents; the assets are ~$1.2 trillion and the capitalized rents ~$17 trillion, so the entire live dispute concerns a rent stream that must grow several-fold and persist. The rents currently earned are predominantly scarcity rents in the supply chain - the mean-reverting kind - and they are simultaneously embedded in the asset book values, so competitive normalization shrinks numerator and denominator together. The revenue that must eventually fund the rents has no sufficient source except labor substitution, at a GDP share that history says takes decades to grow, and the interim is bridged by a circular financing structure that consolidates to a negative external cash flow funded continuously by index capital, debt, and sovereigns - a socialized buildout in real time rather than at the crash. The precedent says the private outcome and the social outcome can diverge completely: telecom's investors lost their capital and society received the substrate of the modern world, including - on the longest accounting - the corpus and the cash engine that produced machine intelligence. Whether this cycle's version bequeaths or evaporates turns on asset lives and on what fraction of capability persists in open, unpossessable form.

The variables worth watching, each tied to a specific failure or falsification mode developed above: the CoWoS supply-demand gap closing through 2026 and memory fabs landing 2027-28 (scarcity-rent normalization; the bullwhip); external AI revenue against the ~$1.7 trillion breakeven floor and the 3-6.5 percent-of-GDP ceiling (the transition from capex-funded to demand-funded rents); cloud AI growth rates sustaining above ~50 percent (the falsifier of the pessimistic case); the debt and private-credit share of buildout funding (equity-diffused versus banked-bubble loss distribution); further intra-loop rent renegotiations of the Microsoft-cap type (margin stacking made visible); RPO and backlog concentration against single counterparties (double-capitalization exposure); the IPO absorption of OpenAI, Anthropic, and Databricks paper (the migration of exposure to the public); and the cadence and capability gap of open-weight releases (the size of the permanent endowment if the funding stops).

Data Provenance

Figures in this report divide into three classes. **Reported:** company disclosures and named third-party research - hyperscaler capex and guidance, Nvidia and TSMC results, Gartner spending series, PitchBook's OpenAI analysis, deal terms as covered by Bloomberg and others. **Estimated by the authors:** the AI-attribution shares of capex, the vintage capital-stock model and all of Tables 1-4, the competitive-cost repricing, current external AI revenue ($300-400B), realized rent composition (~$330-350B/yr), historical IT-share-of-GDP ratios, and the labor-decomposition arithmetic. **Historical literature:** the telecom, railway, and vendor-financing records, cited below. Where a range is given, the range is the claim. Nothing here is investment advice; it is an accounting framework with explicit assumptions, built to be argued with.

Appendix A. Capital Stock Model

Vintage gross capex as in Table 1. Asset mix and straight-line lives: IT equipment (accelerators, servers, networking, memory, storage) 62 percent at 5 years; power and cooling infrastructure 23 percent at 15 years; shell and land 15 percent at 30 years; mid-year convention per vintage. Add-ons at mid-2026, net (low/central/high, $B): upstream fab plant AI-allocated 105/125/145; power and grid 45/65/85; model IP at cost excluding compute 35/55/75. Competitive-cost adjustment: accelerator and networking spend taken as 75 percent of the IT stock, repriced to 45 percent of paid price (competitive-margin replacement); other IT repriced to 75 percent; infrastructure and shell at cost. Rent flows in Table 3 use required flow = rent capitalization times (r - g) for the growing case (r = 8%, g = 3%), times r for the flat case, and times (r + d) with d = 10% for the decaying case. Sensitivity: moving every vintage to its high case and Nvidia's implied IT share to 55 percent raises the central net stock by roughly $200B; the rent capitalization changes by less than 2 percent, because it is dominated by the market-value term.

Appendix B. Revenue Requirement Model

Required after-tax profit = required rent flow (Table 3, $18T attribution) + 8 percent normal return on a projected 2030 net stock of ~$5T (from ~$5.3T big-four capex 2025-2030 plus the rest of the ecosystem, net of depreciation). External revenue = profit / consolidated net margin, shown at 12, 20, and 30 percent. GDP bases: world ~$135T and US ~$35T nominal in 2030 (~5 percent nominal growth from ~$118T and ~$30.5T in 2026, IMF-consistent). Breakeven floor = depreciation on the 2030 stock (~$940B/yr at the Appendix A mix, net-to-gross ~0.77) + operations (~$275B) + tax-adjusted cost of capital (~$480B) = ~$1.70T/yr = 1.3 percent of 2030 world GDP. Labor decomposition: revenue R = capture share times labor cost addressed; at 30 percent capture, R of $4.0/5.5/7.0T implies $13.3/18.3/23.3T addressed = 19/26/33 percent of global labor income (53 percent of world GDP).

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[10] The Atlantic (July 2026), essay on AI valuations and circular investment, as excerpted by the requester; the ouroboros characterization and the PitchBook FCF citation appear there.

[11] Press coverage of OpenAI investor communications (February 2026): infrastructure target reduced from ~$1.4T to ~$600B through 2030; ~$280B projected 2030 revenue from $13.1B in 2025; ~$8B 2025 cash burn; ~$100B round at ~$830-852B valuation.

[12] Nvidia quarterly results, FY2025-FY2027: datacenter revenue $115.2B (FY2025), $193.7B (FY2026), $75.2B (Q1 FY2027, total revenue $81.6B, +85% y/y).

[13] Hyperscaler Q4 2025 and Q1 2026 earnings disclosures: combined 2026 capex guidance ~$725B; Microsoft $190B (including ~$25B attributed to memory and component inflation); Google Cloud +63% y/y; Oracle remaining performance obligations ~half a trillion dollars.

[14] TSMC Q1 2026 results and investor commentary: HPC at 61% of revenue; 66.2% gross margin; 2026 capex $52-56B.

[15] Financial Times and CreditSights compilations of hyperscaler capex, AI shares (~75% for 2026), capital intensity (45-57%; Oracle 86%), and debt issuance ($108B in 2025; ~$1.5T projected); Morgan Stanley and Goldman Sachs capex projections as reported.

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Compiled July 27, 2026, from a working dialogue. Estimates are the authors' and carry the uncertainties stated in the Data Provenance note; reported figures are as of their cited dates and may since have been revised.